HYPERSONICS / CFD

Flow around a cylinder at Mach 6

Local-inclination predictions and inviscid CFD of a circular cylinder at ISA 20 km freestream conditions.

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Separated from the original Hypersonics and Advanced Propulsion report. Equations, result figures and technical text retain their original numbering. Shared setup material is included where needed. No simulation results have been changed.

Table 1: Freestream conditions (ISA 20 km, $\gamma = 1.4$, $C_p = 1004.5\,\mathrm{J/(kg\,K)}$).

Quantity	Symbol	Value	Unit
Static pressure	p_∞	5529	Pa
Density	ρ_∞	0.088	$\mathrm{kg/m^3}$
Temperature	T_∞	216.65	K
Speed of sound	a_∞	≈ 295	m/s
Dynamic viscosity	μ	$\approx 1.42 \times 10^{-5}$	Pa·s

2. Governing Equations and Methods

2.1. Local Inclination Methods

At hypersonic Mach numbers, each surface element is treated as an independent wedge. On the leeward side all three methods set $C_p = 0$.

Modified Newtonian Theory applies the relation:

$$C_p = C_{p,\max} \sin^2\theta, \qquad C_{p,\max} = \frac{2}{\gamma M_\infty^2}\left[\frac{p_{02}}{p_\infty} - 1\right], \tag{1}$$

where p_{02}/p_∞ follows from the Rayleigh pitot formula. At $M_\infty = 6$, $C_{p,\max} \approx 1.831$.

Tangent Wedge Method applies oblique shock relations for the local half-angle θ, solving the θ-β-M relation for β and hence p/p_∞. Beyond the sonic point, Modified Newtonian Theory is used as a fallback.

Shock Expansion Theory starts from an oblique shock at the stagnation point and marches downstream via Prandtl-Meyer expansion:

$$\nu(M_2) = \nu(M_1) + |\Delta\theta|, \qquad \nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}. \tag{2}$$

The Mach 6 cylinder case uses approximately 40,000 cells and rhoCentralFoam with Kurganov-Tadmor flux, van Leer reconstruction, a maximum Courant number of 0.4, and slip walls.

The domain is pear-shaped to align mesh lines with the bow shock. A sonicFoam solution was used to initialize the field.

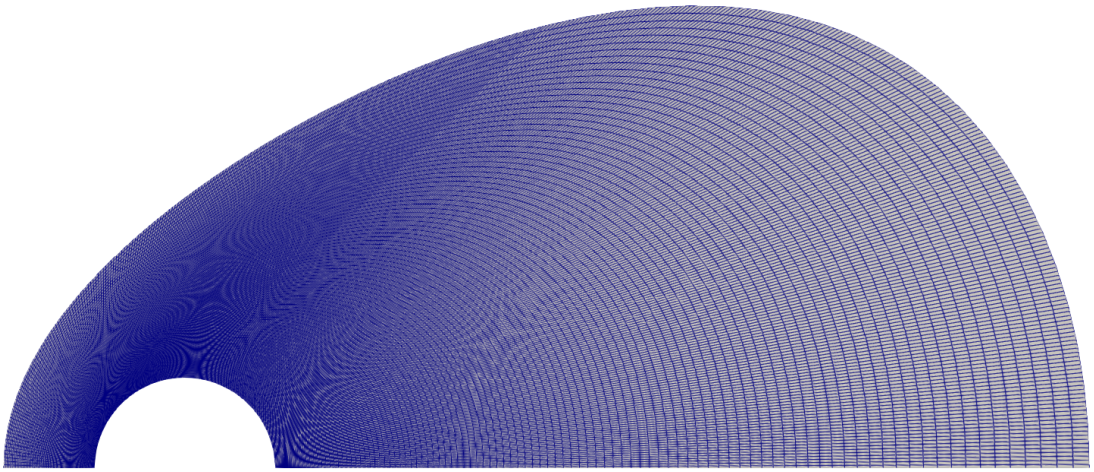


Figure 1: Cylinder mesh with approximately 40,000 cells and symmetry at centreline.

4. Cylinder Results

4.1. Pressure Distribution

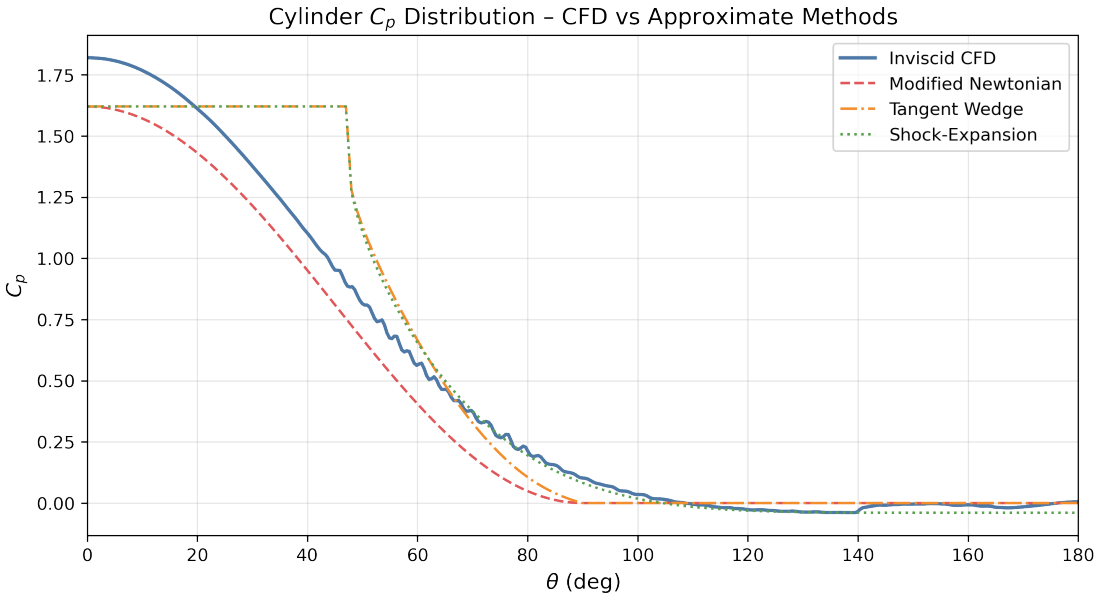


Figure 3: C_p distribution on the cylinder surface compared against inviscid CFD.

Modified Newtonian Theory tracks the C_p of inviscid CFD most uniformly across the windward face. The Tangent Wedge Method and Shock Expansion Theory perform better near the shoulder ($\theta \approx$ 40-70°) where the local geometry more closely resembles a wedge, but both methods struggle at the stagnation point where total-pressure-based C_p is applied up to the sonic point. The CFD peak $C_p \approx 1.8$ exceeds the Modified Newtonian prediction of ≈ 1.62 by roughly 11%. All inclination methods except Shock Expansion Theory collapse to $C_p = 0$ in the shadow region, which is consistent with the assumption of zero pressure perturbation on leeward surfaces. The leeward recirculation visible in Figure 4 is attributed to entropy layer vorticity, described by Crocco's Theorem as

$$\mathbf{V} \times \boldsymbol{\omega} = \nabla \mathcal{H}_0 - T \nabla S \tag{4}$$

where gradients in entropy S across streamlines passing through the curved bow shock drive the rotational flow observed in the wake.

4.2. CFD Results

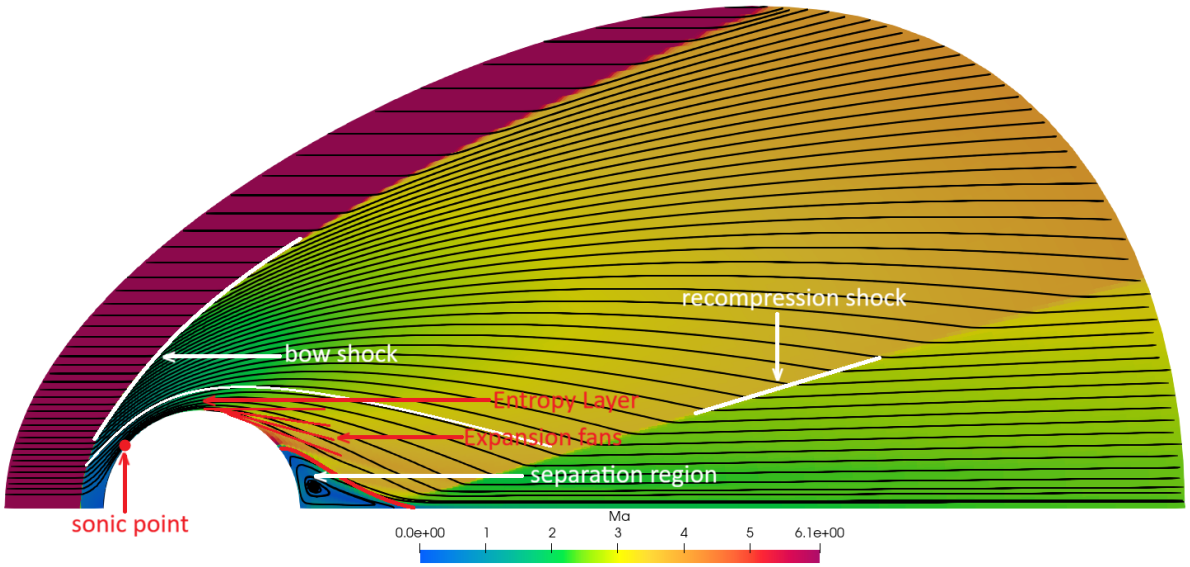
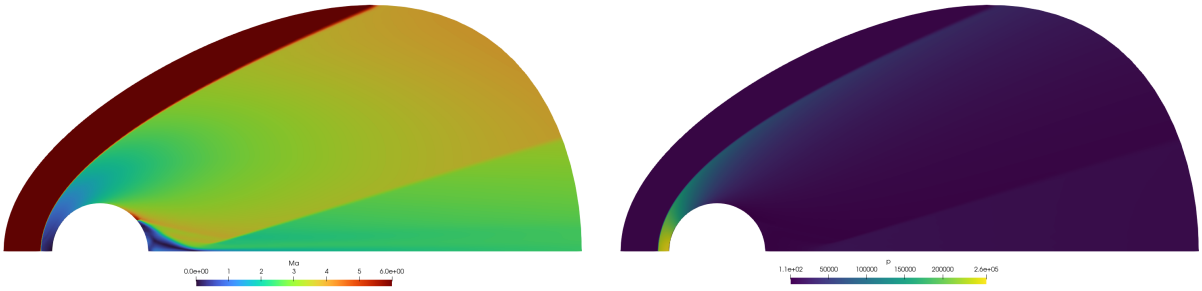


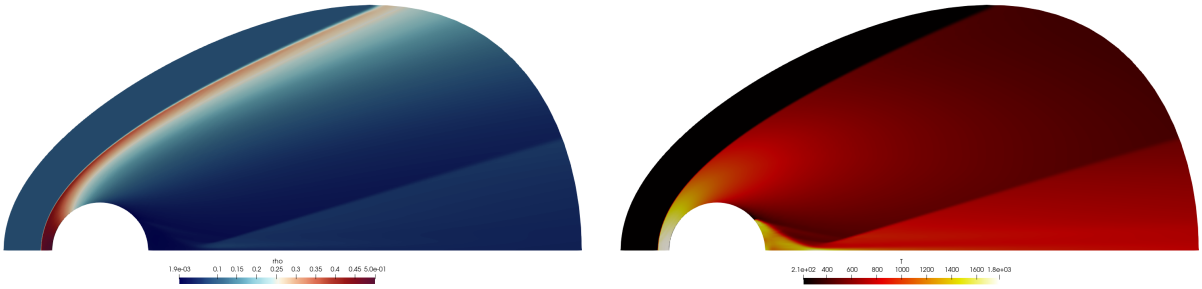
Figure 4: streamlines over the cylinder. Shows recirculation at leeward region possibly effect of entropy layer.



(a) Mach number.

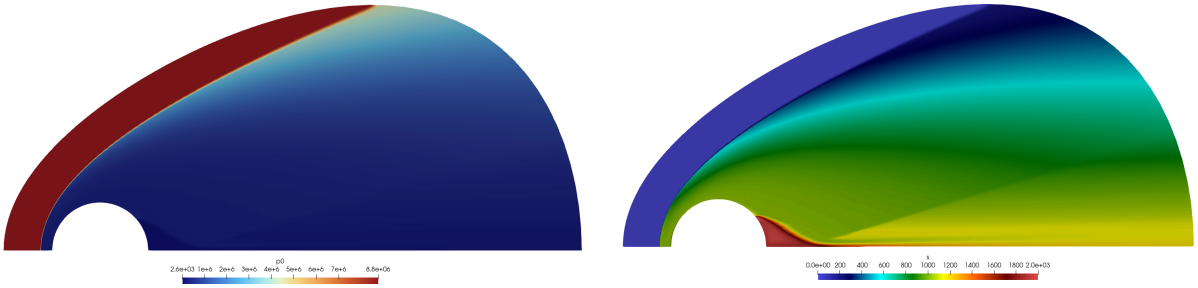
(b) Static pressure.

Figure 5: Cylinder flow field, $M_\infty = 6$.



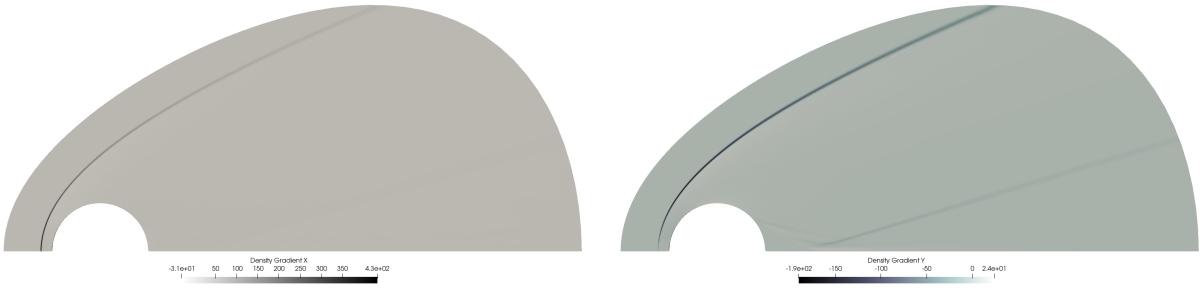
(a) Density. (b) Temperature.

Figure 6: Cylinder density and temperature fields.



(a) Total pressure. (b) Change in Entropy.

Figure 7: Cylinder total pressure and change in entropy.



(a) $\partial\rho/\partial x$. (b) $\partial\rho/\partial y$.

Figure 8: Numerical Schlieren for the cylinder.

Local inclination methods. Modified Newtonian Theory gives the best global agreement because its $\sin^2\theta$ shape inherently captures the blunt-body pressure distribution. The Tangent Wedge Method and Shock Expansion Theory are more accurate near the shoulder ($\theta \approx 40\text{-}70^\circ$) but break down at the stagnation point where the wedge analogy fails. A stagnation point error of roughly 11% is observed.

Appendix: Python Code

Listing 1: Pressure distribution via local inclination methods (`cp_dist.py`).

```
import numpy as np
import pandas as pd
from scipy.optimize import brentq

M = 6.0
gamma = 1.4

theta = np.linspace(0, np.pi, 181)
delta = np.pi / 2 - theta

def tbm(beta):
    return np.arctan(2 / np.tan(beta)
                     * (M**2 * np.sin(beta)**2 - 1)
                     / (M**2 * (gamma + np.cos(2 * beta)) + 2))

betas     = np.linspace(np.arcsin(1 / M) + 1e-6, np.pi / 2 - 1e-6, 1000)
tbm_vals  = tbm(betas)
idx_max   = np.argmax(tbm_vals)
delta_det = tbm_vals[idx_max]
beta_det  = betas[idx_max]

def nu_pm(mach):
    x = max(mach**2 - 1, 0.0)
    return (np.sqrt((gamma + 1) / (gamma - 1))
            * np.arctan(np.sqrt((gamma - 1) / (gamma + 1) * x))
            - np.arctan(np.sqrt(x)))

Cp_stag = 2 / (gamma * M**2) * ((2 * gamma * M**2 - (gamma - 1)) / (gamma + 1) -
    1)

Cp_newt = np.where(delta >= 0, Cp_stag * np.sin(delta)**2, 0.0)

Mn1  = M * np.sin(beta_det)
M1n  = np.sqrt(((gamma - 1) * Mn1**2 + 2) / (2 * gamma * Mn1**2 - (gamma - 1)))
M1   = M1n / np.sin(beta_det - delta_det)
p1   = (2 * gamma * Mn1**2 - (gamma - 1)) / (gamma + 1)
nu1  = nu_pm(M1)

nu_sh = nu1 + delta_det
M_sh  = brentq(lambda mach: nu_pm(mach) - nu_sh, 1.0 + 1e-9, 50.0)
p_sh  = p1 * ((1 + (gamma - 1) / 2 * M1**2)
              / (1 + (gamma - 1) / 2 * M_sh**2)) ** (gamma / (gamma - 1))

Cp_tw = np.zeros_like(theta)
for i, d in enumerate(delta):
    if d >= delta_det:
        Cp_tw[i] = Cp_stag
    elif d > 1e-6:
```

```
        b  = brentq(lambda b: tbm(b) - d, np.arcsin(1 / M) + 1e-9, beta_det - 1e-9)
        Mn = M * np.sin(b)
        Cp_tw[i] = 2 / (gamma * M**2) * ((2 * gamma * Mn**2 - (gamma - 1)) / (gamma + 1) - 1)

Cp_se = np.zeros_like(theta)
for i, d in enumerate(delta):
    if d >= delta_det:
        Cp_se[i] = Cp_stag

    elif d > 1e-9:
        nu2  = nu1 + (delta_det - d)
        M2   = brentq(lambda mach: nu_pm(mach) - nu2, 1.0 + 1e-9, 50.0)
        p2   = p1 * ((1 + (gamma - 1) / 2 * M1**2)
                     / (1 + (gamma - 1) / 2 * M2**2)) ** (gamma / (gamma - 1))
        Cp_se[i] = 2 / (gamma * M**2) * (p2 - 1)

    elif abs(d) < 1e-9:
        Cp_se[i] = 2 / (gamma * M**2) * (p_sh - 1)

    else:
        nu2 = nu_sh + abs(d)
        nu2 = min(nu2, nu_pm(50.0) - 1e-9)
        M2  = brentq(lambda mach: nu_pm(mach) - nu2, M_sh + 1e-9, 50.0)
        p2  = p_sh * ((1 + (gamma - 1) / 2 * M_sh**2)
                      / (1 + (gamma - 1) / 2 * M2**2)) ** (gamma / (gamma - 1))
        Cp_se[i] = 2 / (gamma * M**2) * (p2 - 1)

df = pd.DataFrame({
    "theta_deg":         np.degrees(theta),
    "delta_deg":         np.degrees(delta),
    "Cp_Newtonian":      Cp_newt,
    "Cp_TangentWedge":   Cp_tw,
    "Cp_ShockExpansion": Cp_se,
})
df.to_csv("cylinder_Cp.csv", index=False)
print(df.to_string(index=False))
```